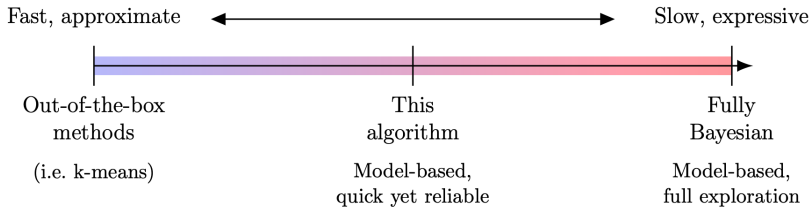


A rapid algorithm for inferring latent mixture structure in replicate social science data

Nathan Kellerman

Bowdoin College · Department of Mathematics





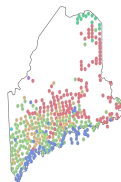
Theory

- ▶ Professor pedagogy toy example
 - ▶ How can we uncover hidden voting populations in data?
- ▶ Political science framing
 - ▶ Can we mathematically model voting ideology in referendum?

$$\log \mathcal{L}(\vec{\theta}; y)$$

Application

- ▶ Simulation studies
 - ▶ How rapid and accurate is the algorithm?
- ▶ Case studies
 - ▶ Maine referendum voting
 - ▶ Switzerland referendum voting (previously inaccessible)



Assume: Bowdoin College has 200 faculty, each with unique membership to one of 33 academic departments

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Data: For each faculty, record binary support for the following:

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Data: For each faculty, record binary support for the following:

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3. All students should pursue undergraduate research

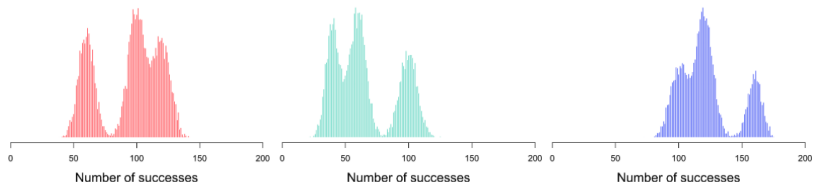
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	$Q_1, \text{ Yes}$	$Q_1, \text{ No}$	$Q_2, \text{ Yes}$	$Q_2, \text{ No}$	$Q_3, \text{ Yes}$	$Q_3, \text{ No}$
Department 1	$y_{1,1}$	$n_{1,1}$	$y_{1,2}$	$n_{1,2}$	$y_{1,3}$	$n_{1,3}$
Department 2	$y_{2,1}$	$n_{2,1}$	$y_{2,2}$	$n_{2,2}$	$y_{2,3}$	$n_{2,3}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Department 33	$y_{33,1}$	$n_{33,1}$	$y_{33,2}$	$n_{33,2}$	$y_{33,3}$	$n_{33,3}$

The shape of professor pedagogy



1. I train my students for a professional workplace
2. Learning should always be difficult
3. All students should pursue undergraduate research

What cultural archetypes of professors exist at Bowdoin?

Three big questions

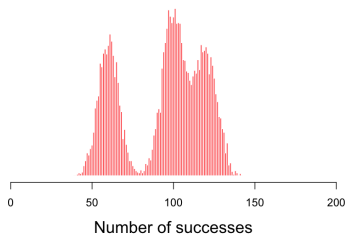
1. How many educational ideologies exist among Bowdoin faculty?
2. How are departments mixed by ideology?
3. What parameters define these ideologies?



- ▶ $i = 1, \dots, M = 33$ departments
- ▶ $q = 1, \dots, Q = 3$ referendum questions
- ▶ y_{iq} yes responses, per department, per question
- ▶ N_{iq} polled faculty per department, per question
- ▶ $k = 1, \dots, K$ latent cultures
- ▶ $z_i \in \{1, \dots, K\}$ latent assignment z per department

A simple mixture distribution

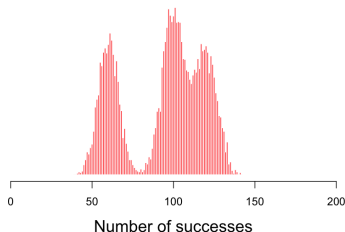
Q1 Support Counts



$$Y \sim \lambda_1 \cdot \pi_1 + \lambda_2 \cdot \pi_2 + \lambda_3 \cdot \pi_3$$

- ▶ $\lambda_1 + \lambda_2 + \lambda_3 = 1$
- ▶ each π_k arises from some well-behaved distributional family $\mathcal{T}(\theta)$

Q1 Support Counts



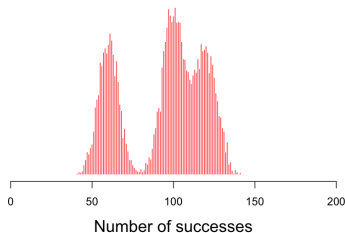
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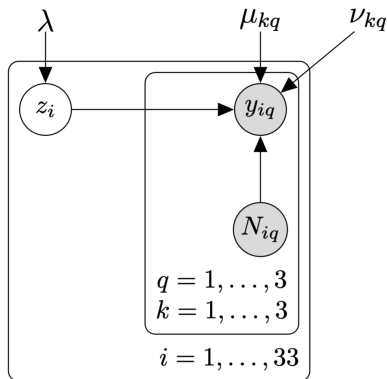
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$$f_k(y_i) = \text{BetaBinomial}(y_i | N_i, \mu_k, \nu_k)$$

Across all $Q = 3$ questions:

$$\pi(\vec{\theta}_k) = \prod_{q=1}^Q f_{kq}(y_{iq})$$



Data likelihood $\mathcal{L}(\vec{\theta}, y)$ across K cultures:

$$\sum_{i=1}^M \log \left\{ \sum_{k=1}^K \lambda_k \left(\prod_{q=1}^Q \text{BetaBinomial}(\mathbf{y}_{iq} | N_{iq}, \mu_{kq}, \nu_{kq}) \right) \right\}$$

Bayesian approach

- ▶ Examine $\mathbb{P}(\theta|\mathcal{D})$
- ▶ Find distributions of parameters θ by conditioning upon the data
- ▶ Explore all parameters

Frequentist approach

- ▶ Examine $\mathbb{P}(\mathcal{D}|\theta)$
- ▶ Construct data likelihood $\mathcal{L}(\vec{\theta}, y)$
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Case Study	# of Locales	# of Questions	Polling Year Range
Pedagogy Toy Example	33	3	NA
Maine Referendum Data	423	70	2008 – 2024
Switzerland Referendum Data	2048	390	1981 – 2026

Rapid algorithm $\equiv$ maximum likelihood estimation procedure

Approach: Try to take derivatives

$$\frac{\partial}{\partial \mu_{kq}} \left[\sum_{i=1}^M \log \left\{ \sum_{k=1}^K \lambda_k \left(\prod_{q=1}^Q \text{BetaBinomial}(\mathbf{y}_{iq} | N_{iq}, \mu_{kq}, \nu_{kq}) \right) \right\} \right]$$

Fact: Finding $\arg \max_{\vec{\theta}} \log \mathcal{L}(\vec{\theta}; X)$ is hard

Let's “pretend” to know $\mathcal{Z}$ and let our data guide us in the right direction

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Consider $\mathcal{D} = (X, \mathcal{Z}) \equiv (\text{observed } X, \text{unobserved latent assignments } \mathcal{Z})$

Definition: the complete data log-likelihood is $\log \mathcal{L}(\vec{\theta}; X, \mathcal{Z})$

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Claim: Finding $\arg \max_{\vec{\theta}} \log \mathcal{L}(\vec{\theta}; X, \mathcal{Z})$ may be easier

Useful definition: The probability that department i is in cluster k is

$$\gamma_{ik} = \mathbb{P}(Z_i = k | x_{iq}, \theta_{t+1}) = \frac{\lambda_k \prod_{q=1}^Q \text{BB}(x_{iq} | N_{iq}, \mu_{kq}, \nu_{kq})}{\sum_{k=1}^K \lambda_k \prod_{q=1}^Q \text{BB}(x_{iq} | N_{iq}, \mu_{jq}, \nu_{jq})}$$

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Amazing fact: $\mathcal{L}(\theta_t; X) \geq \mathcal{Q}(\theta_t, \theta_{t+1})$, for

$$\mathcal{Q}(\theta_t, \theta_{t+1}) = \sum_i \sum_k \mathbb{P}(Z_i = k | x_{iq}, \theta_{t+1}) \log \frac{\mathbb{P}(X_{iq} = x_{iq}, Z_i = k | \theta_t)}{\mathbb{P}(Z_i = k | x_{iq}, \theta_{t+1})}$$

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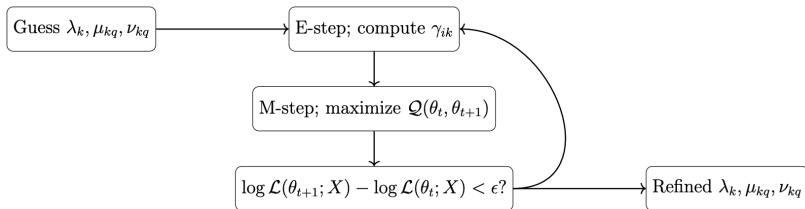
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One EM iteration:

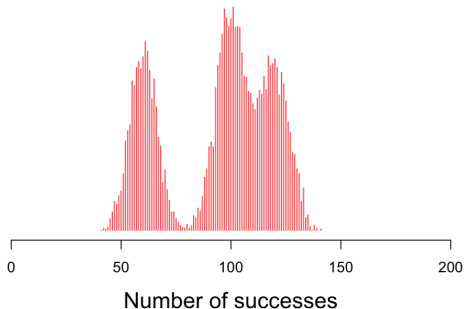
1. E-Step: Compute responsibilities γ_{ik} for each department
2. M-Step: Find the updated parameters which maximize $\mathcal{Q}(\theta_t, \theta_{t+1})$ and thus improve the likelihood $\mathcal{L}(\theta; X)$.

Flow diagram



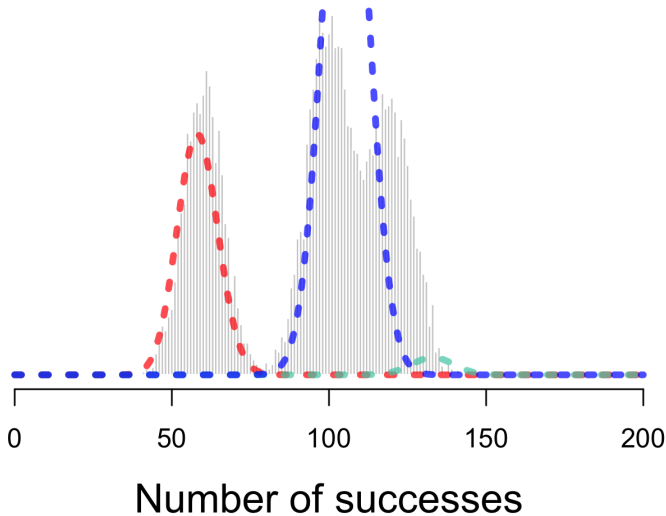
Question 1: I train my students for a professional workplace

Q1 Support Counts

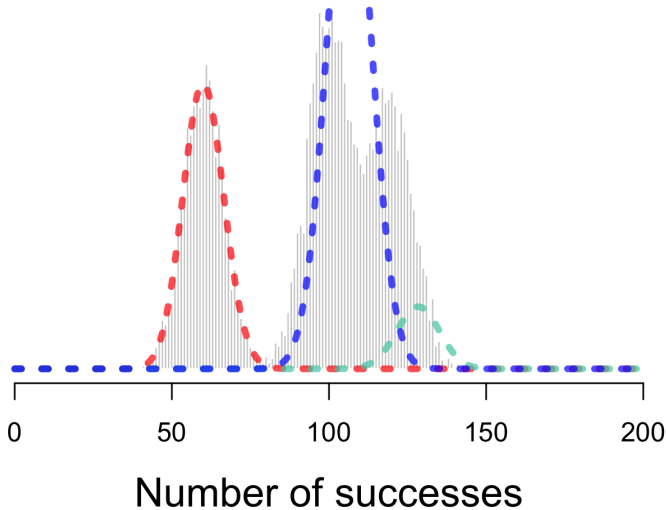


Choose $K = 3$. Guess parameters λ_k , μ_k , and ν_k .

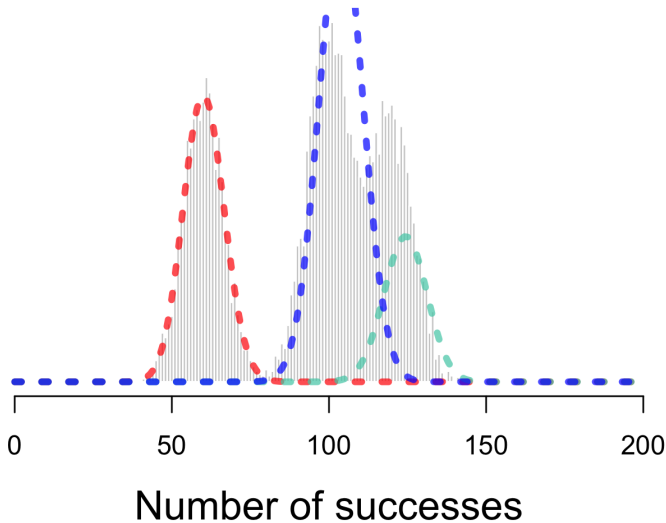
EM Iteration 1



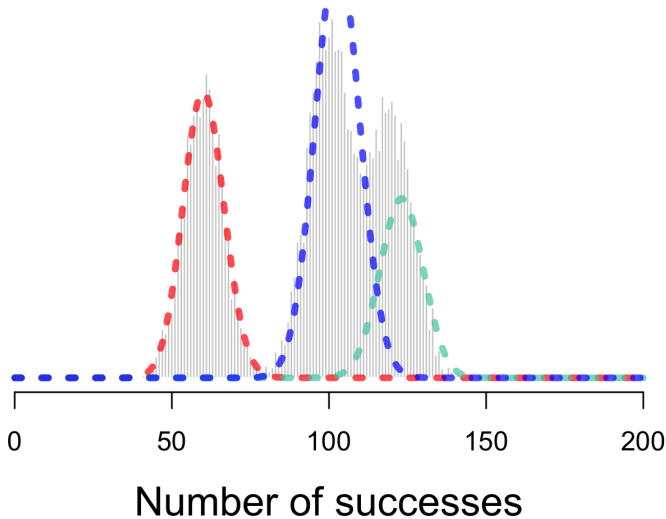
EM Iteration 2



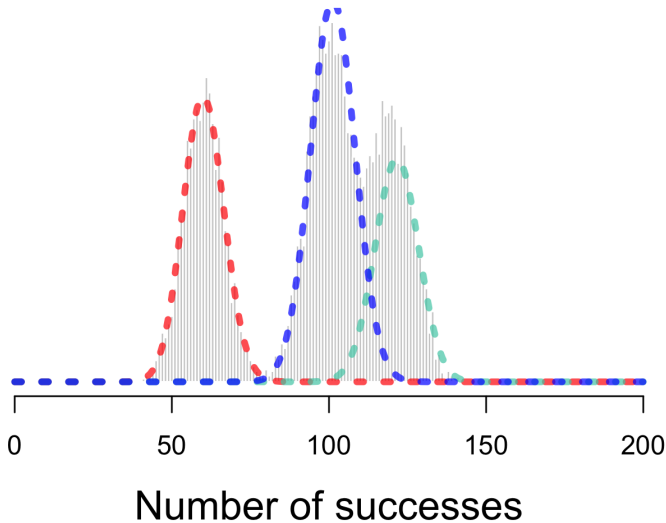
EM Iteration 4



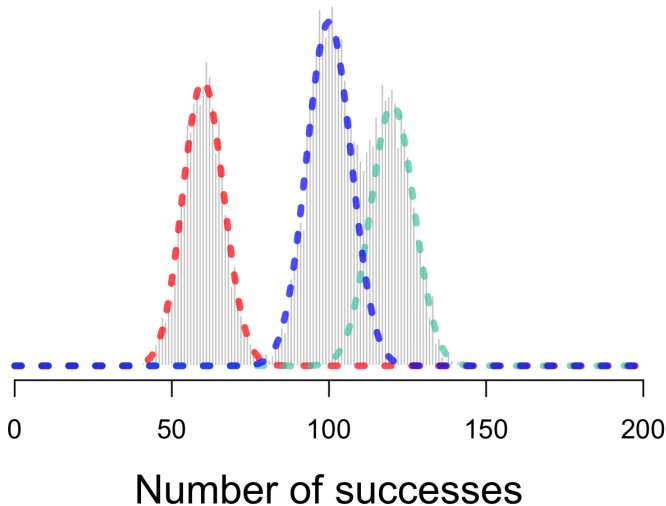
EM Iteration 5



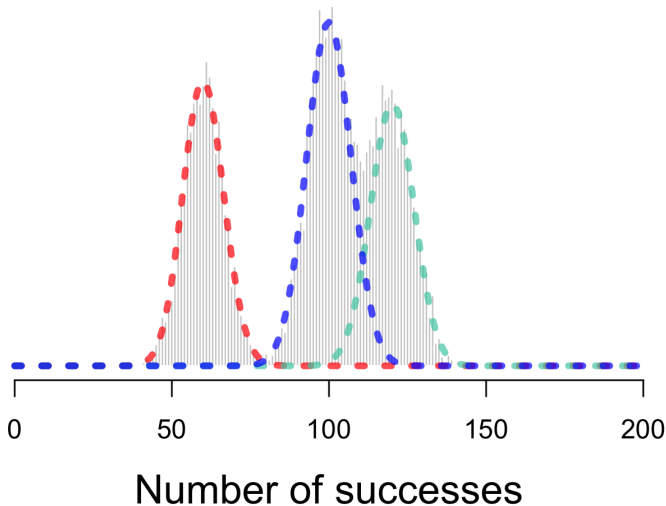
EM Iteration 7



EM Iteration 17



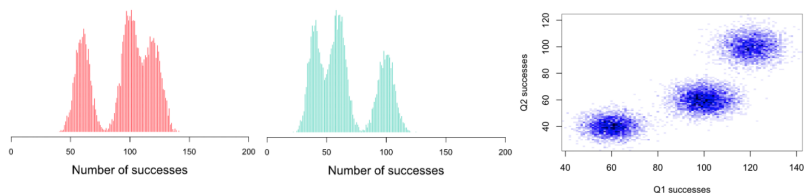
EM Iteration 37



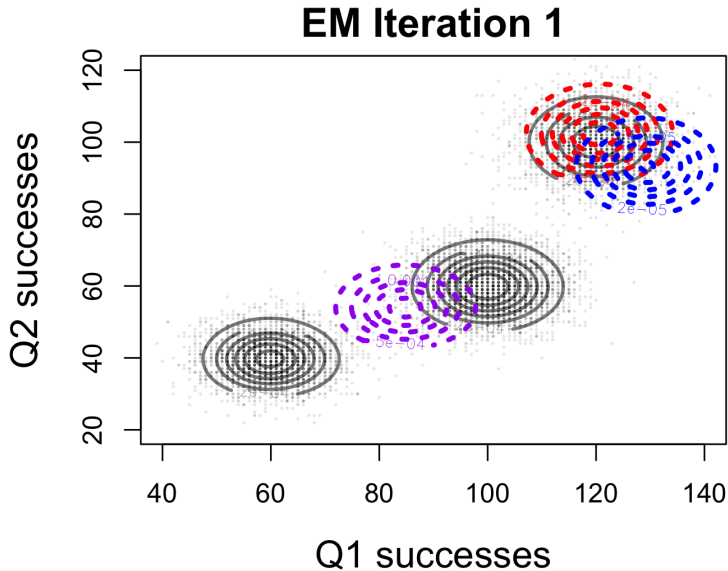
EM over two questions (novel approach)

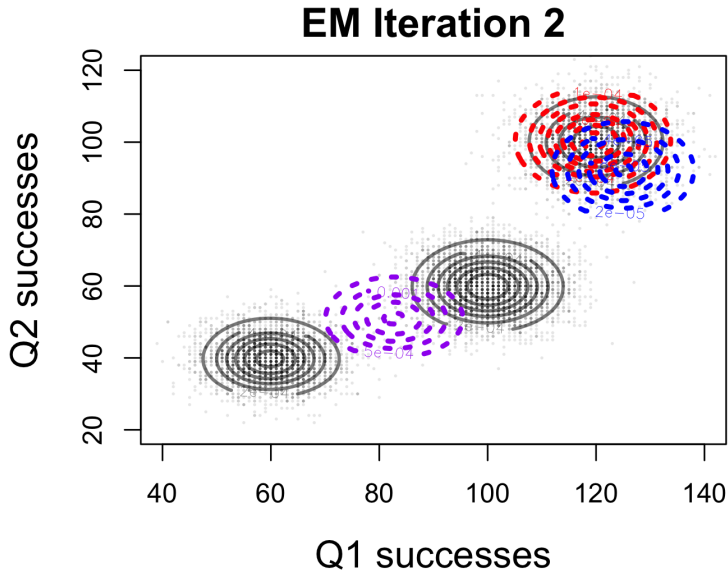
Question 1: I train my students for a professional workplace

Question 2: Learning should always be difficult

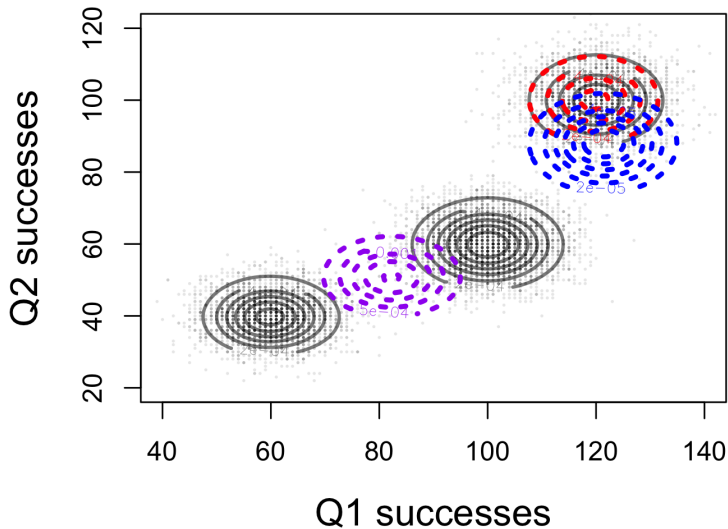


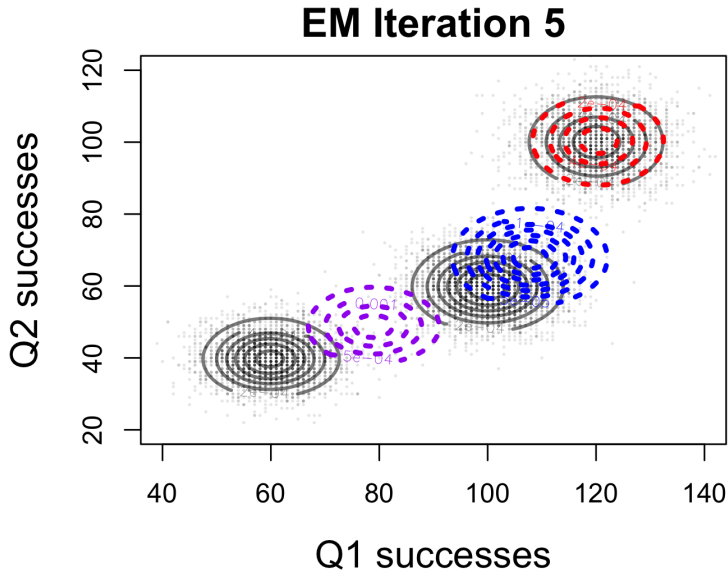
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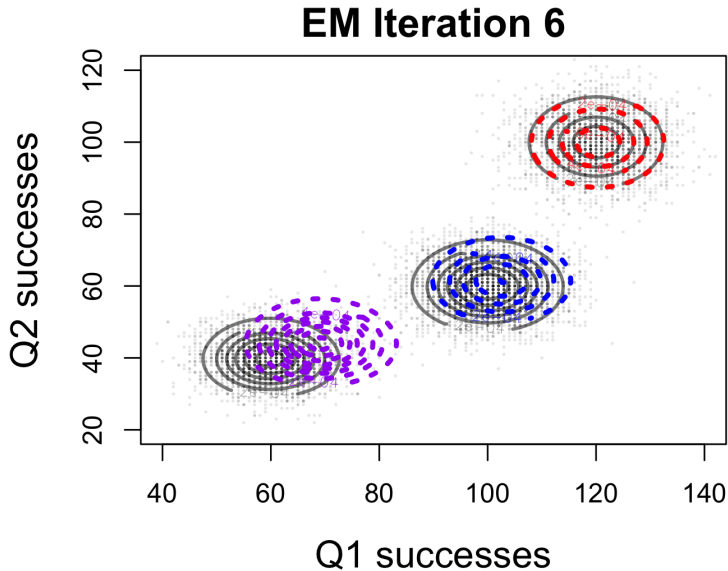




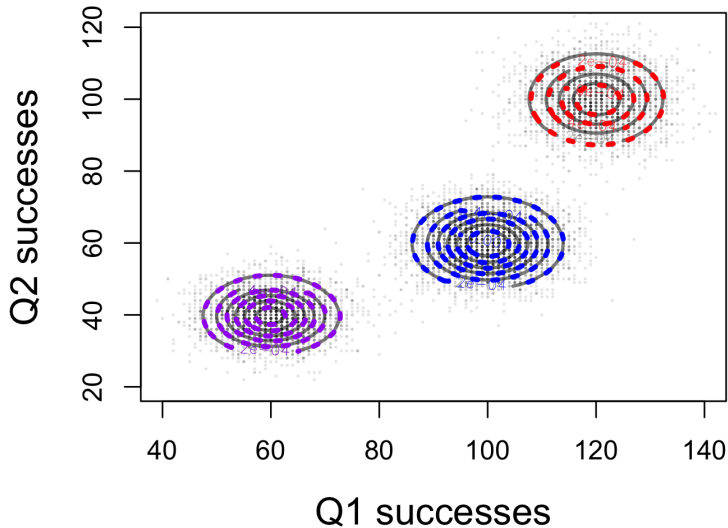
EM Iteration 3







EM Iteration 12



When fed Q questions, the EM converges to local maxima of $\lambda_k, \mu_{kq}, \nu_{kq}$

1. How many educational ideologies exist among Bowdoin faculty?

1.1 K , to be obtained through a model selection heuristic

2. How are departments mixed by ideology?

2.1 Examine responsibilities γ_{ik} , or $\hat{z}_i = \arg \max \gamma_{ik}$

3. What parameters define these ideologies?

3.1 Examine learned μ_{kq}, ν_{kq}

When fed Q questions, the EM converges to local maxima of $\lambda_k, \mu_{kq}, \nu_{kq}$

1. How many educational **political** ideologies exist among Bowdoin faculty **a given electorate**?

1.1 K , to be obtained through a model selection heuristic

2. How are departments **locales** mixed by ideology?

2.1 Examine responsibilities γ_{ik} , or $\hat{z}_i = \arg \max \gamma_{ik}$

3. What parameters define these ideologies?

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Assumptions:

1. Ideologies are latent, consistent sets of principles and beliefs that guide political engagement
 - 1.1 Is the basis for ideology demographic, linguistic, religious, regional, economic?
2. Voting populations are finite mixtures of political cultures, which are informed by ideologies

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Referendum are very direct, not just voting for a political candidate

Referendum ballot measure, Maine 2016

Legalize marijuana for personal use.

3% tax on household income over \$200,000.

Background checks for gun sales and transfers.

Increase minimum wage to \$12 per hour by 2020.

Establish ranked-choice voting.

\$100 million in bonds for transportation projects.

Sanity check: Maine case study

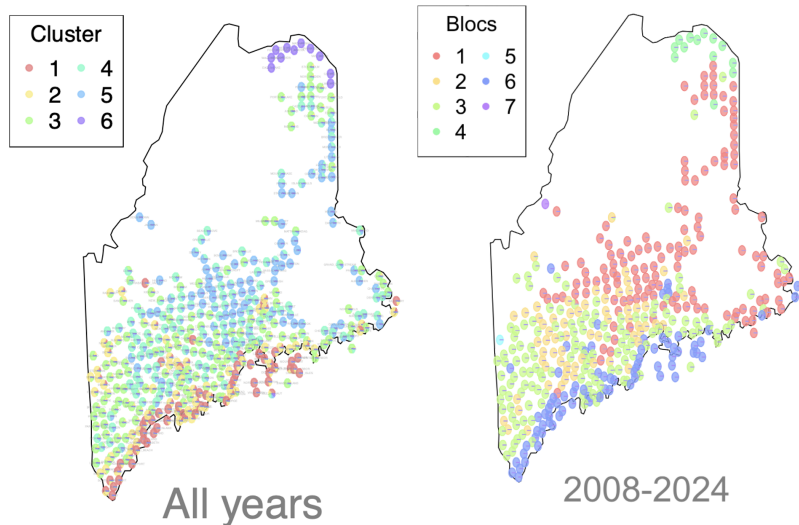


Figure 1: Responsibilities per Maine municipality. Left: Bayesian, 6 weeks of computation. Right: EM, 16.12 seconds of computation.

Important exploratory questions:

1. What is the empirical time complexity of this algorithm?
2. What is the probabilistic accuracy of this inference procedure?

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Creating our own data:

1. Draw a single global mixture weight over the K ideologies
2. Draw a support pattern per latent ideology per question.
3. For each locale, draw a latent ideology label
4. Combine support pattern per ideology with each question, per drawn locale labels

Spanned parameter sets: ^a

$$M \in \{10, 50, 200, 500, 2000\}$$

$$Q \in \{1, 5, 20, 70\}$$

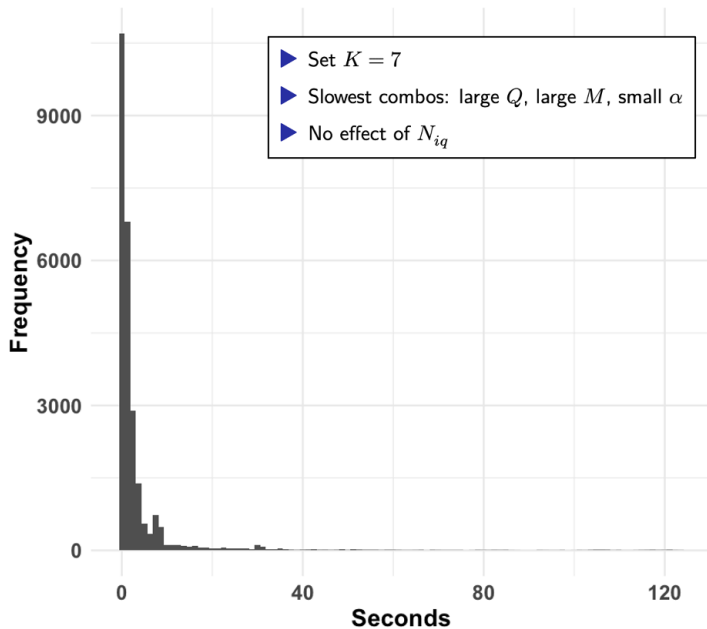
$$N_{iq} \in \{100, 1000, 5000, 30000\}$$

$$\nu_{kq} = 100$$

$$\alpha \in \{0.1, 0.3, 0.5, 0.8\}$$

^a320 unique parameter combos

Computational time simulations



1. Fit model over a range of K
2. Compute model selection heuristic BIC
3. Choose K such that $BIC(K) - BIC(K - 1) < \epsilon$

$$M \in \{10, 50, 200, 500, 2000\}$$

$$Q \in \{1, 5, 20, 70\}$$

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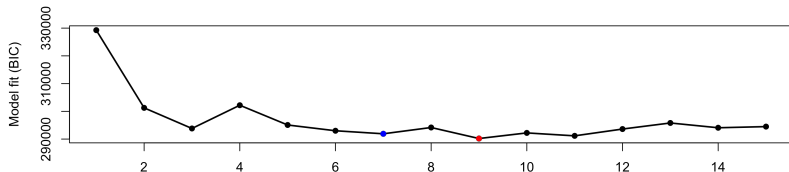
$$\nu_{kq} = 100$$

$$\alpha \in \{0.1, 0.3, 0.5, 0.8\}$$

$$K_{true} \in \{3, \dots, 9\}$$

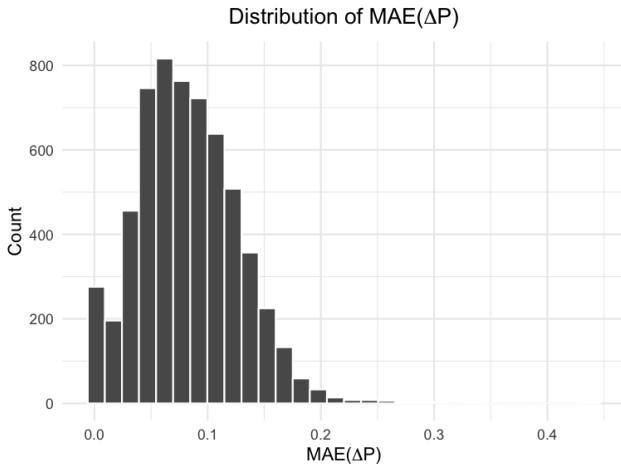
$$K_{search} = 1, \dots, 15$$

$$BIC = \underbrace{\mathcal{K} \ln(M \cdot Q)}_{\text{complexity}} - \underbrace{2 \ln(\mathcal{L})}_{\text{fit}}$$



- ▶ Slowest recovery of 77.5 minutes
 - ▶ $(M, N, Q, \alpha, K) = (2000, 100, 70, 0.5, 3)$
- ▶ Larger Q improves model ability to find K
- ▶ Model over-predicts K in most cases
 - ▶ I.e. Jackman on Maine map
- ▶ Note: 1,680,000 EM runs for this study!

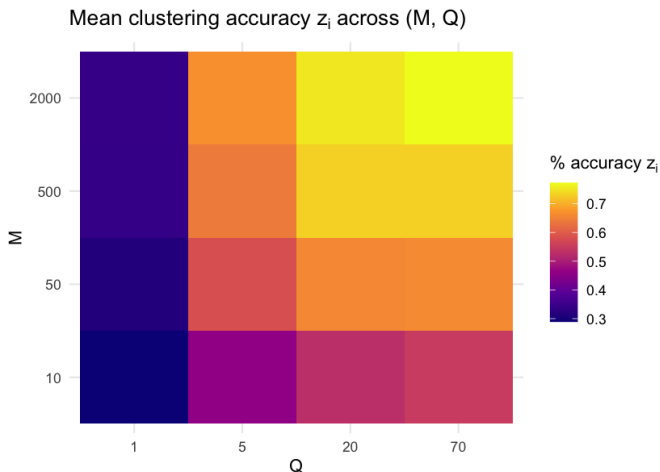
$P_{K \times Q}$ vs $\widehat{P}_{K \times Q}$, across usual parameters with $K \in \{4, \dots, 10\}$



- ▶ On average 8% wrong
- ▶ Q, M have weak effect

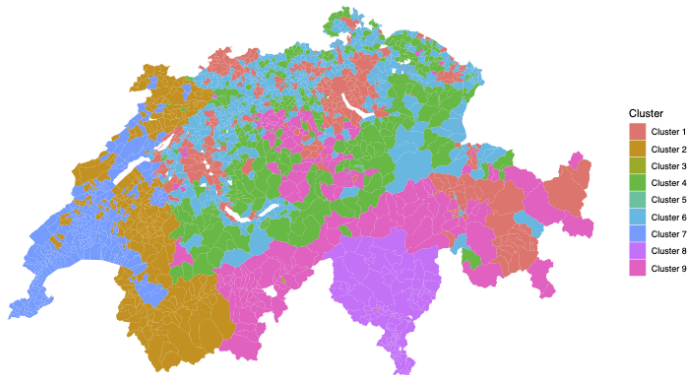
Recovering mixture weights in simulation

z_i vs $\hat{z}_i = \arg \max_k \hat{\gamma}_{ik}$, across usual parameters with $K \in \{4, \dots, 10\}$



► Feature of the algorithm: sparse responsibility vectors

- ▶ Switzerland system of direct democracy
- ▶ Rich longitudinal referendum data
 - ▶ Since 1981, 390 questions across 2048 municipalities (26 cantons)
 - ▶ Bayesian method limits out at ~ 500 municipalities
- ▶ 7.25 minutes to infer model and below hard assignments

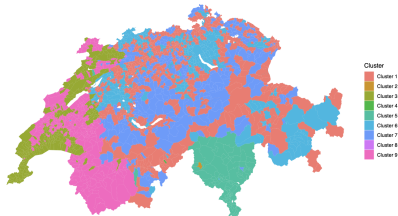


Consistency across 10-year periods

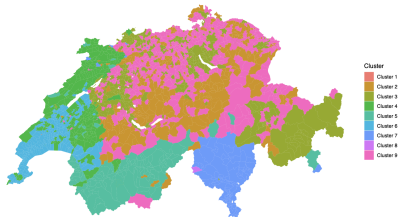
Swiss municipalities: hard cluster assignment (Jan 1981 – Dec 1991)



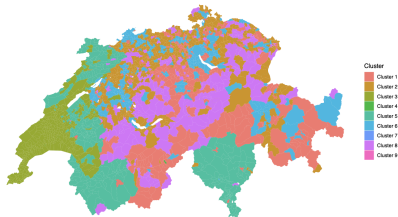
Swiss municipalities: hard cluster assignment (Jan 1992 – Dec 2002)



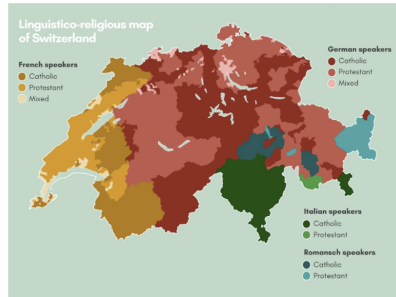
Swiss municipalities: hard cluster assignment (Jan 2003 – Dec 2013)



Swiss municipalities: hard cluster assignment (Jan 2014 – Dec 2026)



Referendum ideology $\iff$ religious and linguistic patterns



- ▶ Religion and language are bases for political ideologies!
 - ▶ Derived from settlement patterns?

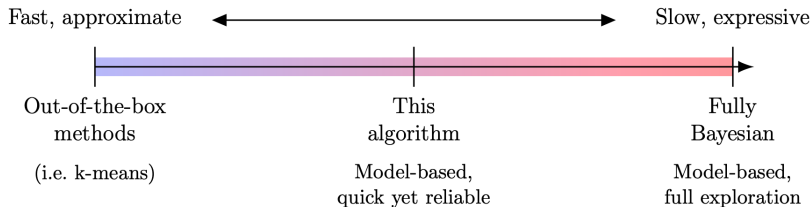
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Can we mathematically model voting ideology in referendum? **Yes!**

- ▶ Religion and language are bases for political ideologies!
- ▶ Derived from settlement patterns?

Can we mathematically model voting ideology in referendum? **Yes!**

Can we pinpoint exactly where ideology comes from? **Not quite...**



A special thanks to Prof. Jack O'Brien and the Mathematics Department



Questions, please!

- ▶ How well does the model handle NA values in data?
 - ▶ Miami referendum data
- ▶ Invariance of method to spatial boundaries?
 - ▶ Canton vs Bezirke vs municipality
- ▶ Methods for more informative exploration of mixture
 - ▶ Laplacian approximation near MLE?